

LR(1) Parsers

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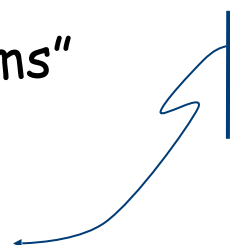
Building LR(1) Tables

How do we build the parse tables for an LR(1) grammar?

- Encode actions & transitions into the ACTION & GOTO tables
- If construction succeeds, the grammar is LR(1)
 - “Succeeds” means defines each table entry uniquely

The Big Picture

- Model the state of the parser with “LR(1) items”
 - The states will be set of LR(1) items
 - Use two functions goto(s, X) and closure(s)
 - goto() tells which state you reach
 - closure() adds information to round out a state
 - Build up the states (sets of LR(1) items) and transitions
 - Use this information to fill in the ACTION and GOTO tables
- fixed-point algorithm



s is a state
X is T or NT

LR(1) Items

We represent a valid configuration of an LR(1) parser with a data structure called an LR(1) item

An LR(1) item is a pair $[P, \delta]$, where

P is a production $A \rightarrow \beta$ with a $\cdot$ at some position in the rhs

δ is a lookahead string of length ≤ 1 (word or EOF)

The $\cdot$ in an item indicates which portion of the righthandside of the production we have seen on the top of the stack

Example: LR(1) Items

Assume the following production

$$A \rightarrow BAg$$

the possible LR(1) items are 4

$[A \rightarrow .BAg, _]$ $[A \rightarrow B.Ag, _]$ $[A \rightarrow BA.g, _]$ $[A \rightarrow BAg., _]$

↓
B = TOP STACK

↓
BA = TOP
STACK

↓
BAg = TOP
STACK

Meaning of an LR(1) Item

$[A \rightarrow \cdot B A g, _]$ means that the input seen so far is consistent with the use of $A \rightarrow B A g$ immediately after the symbol on top of the stack "Possibility"

$[A \rightarrow B \cdot A g, _]$ means that the input seen so far is consistent with the use of $A \rightarrow B A g$ at this point in the parse, and that the parser has already recognized B (that is, B is on top of the stack) "Partially complete"

$[A \rightarrow B A \cdot g, _]$ means that the input seen so far is consistent with the use of $A \rightarrow B A g$ at this point in the parse, and that the parser has already recognized B and A (A is on top of the stack followed by B)

$[A \rightarrow B A g \cdot, \underline{d}]$ means that the parser has seen B A g, and that a lookahead symbol of $\underline{d}$ is consistent with reducing to A "Complete"

LR(1) Items

The production $A \rightarrow B_1 B_2 B_3$ with lookahead $\underline{a}$,
can give rise to 4 items

$[A \rightarrow \cdot B_1 B_2 B_3, \underline{a}]$, $[A \rightarrow B_1 \cdot B_2 B_3, \underline{a}]$, $[A \rightarrow B_1 B_2 \cdot B_3, \underline{a}]$, & $[A \rightarrow B_1 B_2 B_3 \cdot, \underline{a}]$

The set of LR(1) items for a grammar is finite

What's the point of all these lookahead symbols?

- Carry them along to help choosing the correct reduction
- Lookahead are bookkeeping, unless item has $\cdot$ at right end
 - Has no direct use in $[A \rightarrow \beta \cdot \gamma, \underline{a}]$
 - In $[A \rightarrow \beta \cdot, \underline{a}]$, a lookahead of $\underline{a}$ implies a reduction by $A \rightarrow \beta$
 - For a parser state modelled with items $\{ [A \rightarrow \beta \cdot, \underline{a}], [B \rightarrow \gamma \cdot \delta, \underline{b}] \}$,
lookahead of $\underline{a} \Rightarrow$ reduce to A ; lookahead in $\text{FIRST}(\delta) \Rightarrow$ shift

$\Rightarrow$ Limited right context is enough to pick the actions

LR(1) Table Construction

For convenience, we will require that the grammar have an obvious & unique initial symbol — one that does not appear on the rhs of any production.

High-level overview

Build the canonical collection of sets of LR(1) Items

1 Start with an appropriate initial state s_0

- ◆ $[S' \rightarrow \cdot S, \text{EOF}]$ belong to s_0
- ◆ Add to s_0 any equivalent item in the $\text{closure}(s_0)$

A set of LR(1) Items

2 Repeatedly compute, for each state s_k each symbol X , $\text{goto}(s_k, X)$

- ◆ If the set is not already in the collection, add the new state
- ◆ Record all the transitions created by $\text{goto}()$

Terminal and Non-terminal

This eventually reaches a fixed point

Computing Closures

Closure(s) adds all the items implied by the items already in s

- Item $[A \rightarrow \beta \cdot C \delta, \underline{a}]$ in s implies $[C \rightarrow \cdot \tau, \underline{x}]$ for each production with C on the lhs, and each $x \in \text{FIRST}(\delta \underline{a})$
- Since $\beta C \delta$ is a valid rewriting, any way to derive $\beta C \delta$ is a valid rewriting, too

The algorithm

```
Closure( s )  
while ( s is still changing )  
   $\forall$  items  $[A \rightarrow \beta \cdot C \delta, \underline{a}] \in s$   
     $\forall$  productions  $C \rightarrow \tau \in P$   
       $\forall \underline{x} \in \text{FIRST}(\delta \underline{a})$  //  $\delta$  might be  $\epsilon$   
        if  $[C \rightarrow \cdot \tau, \underline{x}] \notin s$   
          then  $s \leftarrow s \cup \{ [C \rightarrow \cdot \tau, \underline{x}] \}$ 
```

- Classic fixed-point method
- Halts because $s \subset \text{ITEMS}$
- Closure "fills out" a state

Lookaheads are
generated here

Example From SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

Initial step builds the item $[Goal \rightarrow \bullet SheepNoise, EOF]$
and takes its closure()

Closure($[Goal \rightarrow \bullet SheepNoise, \underline{EOF}]$)

#	Item	Derived from ...
1	$[Goal \rightarrow \bullet SheepNoise, \underline{EOF}]$	Original item
2	$[SheepNoise \rightarrow \bullet SheepNoise \underline{baa}, \underline{EOF}]$	1, $\delta \underline{a}$ is <u>EOF</u>
3	$[SheepNoise \rightarrow \bullet \underline{baa}, \underline{EOF}]$	1, $\delta \underline{a}$ is <u>EOF</u>
4	$[SheepNoise \rightarrow \bullet SheepNoise \underline{baa}, \underline{baa}]$	2, $\delta \underline{a}$ is <u>baa</u> <u>EOF</u>
5	$[SheepNoise \rightarrow \bullet \underline{baa}, \underline{baa}]$	2, $\delta \underline{a}$ is <u>baa</u> <u>EOF</u>

stop!

4 $\delta \alpha$ baa baa

S_0 (the first state) is

{ $[Goal \rightarrow \bullet SheepNoise, \underline{EOF}]$, $[SheepNoise \rightarrow \bullet SheepNoise \underline{baa}, \underline{EOF}]$,
 $[SheepNoise \rightarrow \bullet \underline{baa}, \underline{EOF}]$, $[SheepNoise \rightarrow \bullet SheepNoise \underline{baa}, \underline{baa}]$,
 $[SheepNoise \rightarrow \bullet \underline{baa}, \underline{baa}]$ }

Computing Gotos

Goto(s,x) computes the state that the parser would reach if it recognized an X while in state s

- Goto({ [A→β•X δ, a] }, X) produces [A→βX •δ, a] (obviously)
- It finds all such items & uses closure() to fill out the state

The algorithm *sposta • di uno in avanti*

```
Goto( s, X )  
  new ← ∅  
  ∀ items [A→β•X δ, a] ∈ s  
    new ← new ∪ { [A→βX •δ, a] }  
  return closure(new)
```

- Not a fixed-point method!
- Straightforward computation
- Uses closure ()

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u> EOF
2			<u>baa</u>

S_0 is { [Goal→ • SheepNoise, EOF], [SheepNoise→ • SheepNoise baa, EOF],
[SheepNoise→ • baa, EOF], [SheepNoise→ • SheepNoise baa, baa],
[SheepNoise→ • baa, baa] }

Goto(S_0 , baa)

- Loop produces

Item	Source
[<u>SheepNoise</u> → <u>baa</u> •, <u>EOF</u>]	Item 3 in s_0
[<u>SheepNoise</u> → <u>baa</u> •, <u>baa</u>]	Item 5 in s_0

- Closure adds nothing since • is at end of rhs in each item

Building the Canonical Collection : The algorithm

```
 $s_0 \leftarrow \text{closure}([S' \rightarrow \cdot S, \underline{\text{EOF}}])$   
 $S \leftarrow \{s_0\}$   
 $k \leftarrow 1$   
while (S is still changing)  
   $\forall s_j \in S \text{ and } \forall x \in (T \cup NT)$   
     $t \leftarrow \text{goto}(s_j, x)$   
    if  $t \notin S$  then  
      name closure( $t$ ) as  $s_k$   
       $S \leftarrow S \cup \{s_k\}$   
      record  $s_j \rightarrow s_k$  on  $x$   
       $k \leftarrow k + 1$   
    else  
       $t$  is  $s_m \in S$   
      record  $s_j \rightarrow s_m$  on  $x$ 
```

Start from $s_0 = \text{closure}([S' \rightarrow \cdot S, \underline{\text{EOF}}])$

Repeatedly construct new states, until all are found

- Fixed-point computation
- Loop adds to S
- $S \subseteq 2^{\text{ITEMS}}$, so S is finite

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

Starts with S_0

$S_0 : \{ [Goal \rightarrow \cdot SheepNoise, \underline{EOF}], [SheepNoise \rightarrow \cdot SheepNoise \underline{baa}, \underline{EOF}],$
 $[SheepNoise \rightarrow \cdot \underline{baa}, \underline{EOF}], [SheepNoise \rightarrow \cdot SheepNoise \underline{baa}, \underline{baa}],$
 $[SheepNoise \rightarrow \cdot \underline{baa}, \underline{baa}] \}$

Iteration 1 computes

$S_1 = Goto(S_0, SheepNoise) =$

$\{ [Goal \rightarrow SheepNoise \cdot, \underline{EOF}], [SheepNoise \rightarrow SheepNoise \cdot \underline{baa}, \underline{EOF}],$
 $[SheepNoise \rightarrow SheepNoise \cdot \underline{baa}, \underline{baa}] \}$

No more for closure!

$S_2 = Goto(S_0, \underline{baa}) = \{ [SheepNoise \rightarrow \underline{baa} \cdot, \underline{EOF}],$
 $[SheepNoise \rightarrow \underline{baa} \cdot, \underline{baa}] \}$

No more for closure!

Iteration 2 computes

$S_3 = Goto(S_1, \underline{baa}) = \{ [SheepNoise \rightarrow SheepNoise \underline{baa} \cdot, \underline{EOF}],$
 $[SheepNoise \rightarrow SheepNoise \underline{baa} \cdot, \underline{baa}] \}$

No more for closure!

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [Goal \rightarrow \cdot SheepNoise, \underline{EOF}], [SheepNoise \rightarrow \cdot SheepNoise \underline{baa}, \underline{EOF}],$
 $[SheepNoise \rightarrow \cdot \underline{baa}, \underline{EOF}], [SheepNoise \rightarrow \cdot SheepNoise \underline{baa}, \underline{baa}],$
 $[SheepNoise \rightarrow \cdot \underline{baa}, \underline{baa}] \}$

$S_1 = Goto(S_0, SheepNoise) =$
 $\{ [Goal \rightarrow SheepNoise \cdot, \underline{EOF}], [SheepNoise \rightarrow SheepNoise \cdot \underline{baa}, \underline{EOF}],$
 $[SheepNoise \rightarrow SheepNoise \cdot \underline{baa}, \underline{baa}] \}$

$S_2 = Goto(S_0, \underline{baa}) = \{ [SheepNoise \rightarrow \underline{baa} \cdot, \underline{EOF}],$
 $[SheepNoise \rightarrow \underline{baa} \cdot, \underline{baa}] \}$

$S_3 = Goto(S_1, \underline{baa}) = \{ [SheepNoise \rightarrow SheepNoise \underline{baa} \cdot, \underline{EOF}],$
 $[SheepNoise \rightarrow SheepNoise \underline{baa} \cdot, \underline{baa}] \}$

Filling in the ACTION and GOTO Tables

The algorithm

x is the state number

$\forall$ set $S_x \in S$

$\forall$ item $i \in S_x$

case 1

if i is $[A \rightarrow \beta \cdot \underline{a} \delta, \underline{b}]$ and $\text{goto}(S_x, \underline{a}) = S_k, \underline{a} \in T$
then $\text{ACTION}[x, \underline{a}] \leftarrow \text{"shift } k\text{"}$

• before terminal
 $\Rightarrow$ shift

case 2

else if i is $[S' \rightarrow S \cdot, \underline{\text{EOF}}]$
then $\text{ACTION}[x, \underline{\text{EOF}}] \leftarrow \text{"accept"}$

have accept

case 3

else if i is $[A \rightarrow \beta \cdot, \underline{a}]$
then $\text{ACTION}[x, \underline{a}] \leftarrow \text{"reduce } A \rightarrow \beta\text{"}$

• at end $\Rightarrow$ reduce

$\forall n \in NT$

if $\text{goto}(S_x, n) = S_k$
then $\text{GOTO}[x, n] \leftarrow k$

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

• before terminal $\Rightarrow$ shift **(k)**

$S_1 = \text{Goto}(S_0, \text{SheepNoise}) =$

$\{ [\text{Goal} \rightarrow \text{SheepNoise } \cdot, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

so, ACTION[$s_0, \underline{\text{baa}}$] is
 "shift S_2 " (case 1)

$S_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_1 = \text{Goto}(S_0, \text{SheepNoise}) =$

$\{ [\text{Goal} \rightarrow \text{SheepNoise } \cdot, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

$S_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

so, ACTION[$S_1, \underline{\text{baa}}$] is
"shift S_3 " (case 1)

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_1 = \text{Goto}(S_0, \text{SheepNoise}) =$

$\{ [\text{Goal} \rightarrow \text{SheepNoise } \cdot, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

so, ACTION[S_1, EOF]
is "accept" (case 2)

$S_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

$S_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_1 = \text{Goto}(S_0, \text{SheepNoise}) =$

$\{ [\text{Goal} \rightarrow \text{SheepNoise } \cdot, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

so, ACTION[S_2, EOF] is
 "reduce 2" (case 3)

$S_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

ACTION[$S_2, \underline{\text{baa}}$] is
 "reduce 2" (case 3)

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

$S_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_1 = \text{Goto}(S_0, \text{SheepNoise}) =$

ACTION[$S_3, \underline{\text{EOF}}$] is
 "reduce 1" (case 3)

$\underline{\text{EOF}}, [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $\text{Noise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$S_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

$S_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

ACTION[$S_3, \underline{\text{baa}}$] is
 "reduce 1", as well

Example from SheepNoise

0	Goal	→	SheepNoise
1	SheepNoise	→	SheepNoise <u>baa</u>
2			<u>baa</u>

The *GOTO* Table records *Goto* transitions on NTs

$s_0 : \{ [\text{Goal} \rightarrow \cdot \text{SheepNoise}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \cdot \text{SheepNoise } \underline{\text{baa}}, \underline{\text{baa}}],$
 $[\text{SheepNoise} \rightarrow \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$s_1 = \text{Goto}(S_0, \text{SheepNoise}) =$ | Puts s_1 in $\text{GOTO}[s_0, \text{SheepNoise}]$

$\{ [\text{Goal} \rightarrow \text{SheepNoise } \cdot, \underline{\text{EOF}}], [\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \cdot \underline{\text{baa}}, \underline{\text{baa}}] \}$

$s_2 = \text{Goto}(S_0, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$ Based on T, not NT and written into the ACTION table

$s_3 = \text{Goto}(S_1, \underline{\text{baa}}) = \{ [\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{EOF}}],$
 $[\text{SheepNoise} \rightarrow \text{SheepNoise } \underline{\text{baa}} \cdot, \underline{\text{baa}}] \}$

Only 1 transition in the entire *GOTO* table

Remember, we recorded these so we don't need to recompute them.

ACTION & GOTO Tables

0	<i>Goal</i>	→	<i>SheepNoise</i>
1	<i>SheepNoise</i>	→	<i>SheepNoise</i> <u>baa</u>
2			<u>baa</u>

Here are the tables for the augmented left-recursive SheepNoise grammar

The tables

ACTION TABLE		
State	EOF	<u>baa</u>
0	—	<i>shift 2</i>
1	<i>accept</i>	<i>shift 3</i>
2	<i>reduce 2</i>	<i>reduce 2</i>
3	<i>reduce 1</i>	<i>reduce 1</i>

GOTO TABLE	
State	<i>SheepNoise</i>
0	1
1	0
2	0
3	0

Note that this is the left-recursive SheepNoise; the book shows the right-recursive version.

What can go wrong?

What if set s contains $[A \rightarrow \beta \cdot \underline{a} \gamma, \underline{b}]$ and $[B \rightarrow \beta \cdot, \underline{a}]$?

- First item generates "shift", second generates "reduce"
- Both define $ACTION[s, \underline{a}]$ — cannot do both actions
- This is a fundamental ambiguity, called a shift/reduce error
- Modify the grammar to eliminate it (if-then-else)
- Shifting will often resolve it correctly

What if set s contains $[A \rightarrow \gamma \cdot, \underline{a}]$ and $[B \rightarrow \gamma \cdot, \underline{a}]$?

- Each generates "reduce", but with a different production
- Both define $ACTION[s, \underline{a}]$ — cannot do both reductions
- This is a fundamental ambiguity, called a reduce/reduce conflict
- Modify the grammar to eliminate it

In either case, the grammar is not LR(1)

LR(k) versus LL(k)

Finding Reductions

LR(k) $\Rightarrow$ Each reduction in the parse is detectable with

- $\rightarrow$ the complete left context,
- $\rightarrow$ the reducible phrase, itself, and
- $\rightarrow$ the k terminal symbols to its right

generalizations of
LR(1) and LL(1) to
longer lookaheads

LL(k) $\Rightarrow$ Parser must select the reduction based on

- $\rightarrow$ The complete left context
- $\rightarrow$ The next k terminals

Thus, LR(k) examines more context

Non-LL(k) Grammars

0	$B \rightarrow R$
1	$\quad (B)$
2	$R \rightarrow E = E$
3	$E \rightarrow \underline{a}$
4	$\quad \underline{b}$
5	$\quad (E + E)$

Example from D.E Knuth, "Top-Down Syntactic Analysis," Acta Informatica, 1:2 (1971), pages 79-110

This grammar is actually LR(0)

0	$S \rightarrow \underline{a} A \underline{b}$
1	$\quad \underline{c}$
2	$A \rightarrow \underline{b} S$
3	$\quad B \underline{b}$
4	$B \quad \underline{a} A$
5	$\quad \underline{c}$

Example from Lewis, Rosenkrantz, & Stearns book, "Compiler Design Theory," (1976), Figure 13.1

Summary

	Advantages	Disadvantages
Top-down Recursive descent, LL(1)	Fast Good locality Simplicity Good error detection	Hand-coded High maintenance Right associativity
LR(1)	Fast Deterministic langs. Automatable Left associativity	Large working sets Poor error messages Large table sizes

Exercise

Consider the following grammar:

$$\begin{aligned} \textit{Start} &\rightarrow S \\ S &\rightarrow A a \\ A &\rightarrow B C \\ &\quad | B C f \\ B &\rightarrow b \\ C &\rightarrow c \end{aligned}$$

- Construct the canonical collection of sets of LR(1) items for this grammar.
- Derive the Action and Goto tables.
- Is the grammar LR(1)?

Parse the string bcfa and the string bca

$s_0:$
 $\text{start} \rightarrow \cdot S, \text{EOF}$

$S \rightarrow \cdot A a, \text{EOF}$

$A \rightarrow \cdot BC, a$

$A \rightarrow \cdot BCF, a$

$B \rightarrow \cdot b, c$

$s_1: \text{goto}(s_0, S)$

$\text{start} \rightarrow S \cdot, \text{EOF (accept)}$

$s_2: \text{goto}(s_0, A)$

$S \rightarrow A \cdot a, \text{EOF}$

$s_3: \text{goto}(s_0, B)$

$A \rightarrow B \cdot C, a$

$A \rightarrow B \cdot CF, a$

$C \rightarrow \cdot c, a$

$C \rightarrow \cdot c, f$

$s_4: \text{goto}(s_0, b)$

$B \rightarrow b \cdot, c$

$s_5: \text{goto}(s_2, a)$

Exercise

Construct the table for descendent parser, for the language defined by the following grammar:

$P \rightarrow \text{begin } L \text{ end}$

$L \rightarrow ST$

$T \rightarrow ST \mid \varepsilon$

$S \rightarrow \text{id} := E; \mid \text{read}(\text{id}); \mid \text{write}(E);$

$E \rightarrow FG$

$G \rightarrow + FG \mid \varepsilon$

$F \rightarrow (E) \mid \text{id}$